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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Hotel Room Reservation and Guest Management System

Submitted by

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25BCS026

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DECLARATION

I hereby declare that the project entitled "**Hotel Room Reservation and Guest Management System"** submitted in partial fulfillment of the requirements for the course Java Programming is a **record of my original work** carried out under the guidance of **DR S BOOPALAN**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Jarvis Steven J

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CERTIFICATE

This is to certify that the project entitled **"Hotel Room Reservation and Guest Management System"** is a **bona fide work** carried out by **Jarvis Steven J, 1st Bsc CS 'A'**, B.Sc.Computer Science, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date: 10.4.2026

Submitted for the Viva-Voce Examination held on

_____10.4.2026_____

Guide:
(Technical Trainer, KGCAS)

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Head of the Department(Technical Head, KGCAS)

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ABSTRACT

The **Hotel Room Reservation and Guest Management System** is a JavaFX-based application designed to automate the process of room booking and guest management in hotels. In many small hotels, reservations are handled manually, which can lead to errors such as double booking, poor record management, and difficulty in tracking guest details.

This project provides a simple solution where users can enter guest details, reserve rooms, view current reservations, and perform check-out operations. The system ensures that a room cannot be booked by more than one guest at the same time by implementing a validation check.

The application is developed using Java and JavaFX, applying concepts such as event handling, loops, conditional statements, and data structures like ArrayList. The system provides a user-friendly interface and displays reservation details clearly.

The main objective of the project is to improve efficiency, reduce manual errors, and demonstrate the practical use of Java programming in developing real-world applications.

1. INTRODUCTION

In the hospitality industry, managing room reservations efficiently is important for smooth operations and customer satisfaction. In many small hotels, room booking is done manually, which can lead to problems such as double booking, confusion in records, and difficulty in tracking guest information.

Manual systems are time-consuming and prone to human errors. Staff may accidentally assign the same room to multiple guests or fail to update records properly during check-out.

With the advancement of technology, automated reservation systems have become essential. A computerized system can manage bookings accurately, prevent conflicts, and maintain proper records.

The Hotel Room Reservation and Guest Management System is developed using JavaFX to provide a simple and effective solution for managing hotel bookings. The system allows users to enter guest details, reserve rooms, check room availability, and perform check-out operations.

This project demonstrates the use of Java programming and GUI development in solving real-world problems in hotel management.

1.1 Objective

- To develop a JavaFX-based hotel reservation system
- To automate room booking and guest management
- To prevent double booking of rooms
- To manage guest check-in and check-out efficiently
- To apply programming concepts such as loops, conditions, and event handling

1.2 Overview

The system is a simple GUI-based application developed using JavaFX. It allows users to input guest details and assign rooms.

The system checks whether a room is already booked before confirming a reservation. It also allows users to view all current bookings and perform check-out operations.

All data is stored temporarily using ArrayList, and the system updates information dynamically based on user actions.

1.3 Features

The **Hotel Room Reservation and Guest Management System** includes the following main features:

- Room Reservation
- View Reservations
- Prevent Double Booking
- Guest Check-Out
- Simple User Interface

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

In many small hotels, room reservations are managed manually, leading to issues such as:

- Double booking of rooms
- Poor record management
- Time-consuming operations
- Difficulty in tracking guest details

This creates inefficiency and confusion in hotel operations.

2.2 Project Overview

The system provides a digital platform to manage hotel reservations efficiently.

It includes:

- Input of guest details
- Room allocation
- Validation of room availability
- Display of reservation details
- Check-out functionality

2.3 Scopes and Limitations

Scope:

- **Small hotels**
- **Guest reservation management**
- **Demonstration of JavaFX concepts**

Limitations:

- **No database (temporary storage only)**
- **Basic UI**
- **Single-user system**

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

Input Module – Takes guest name and room number

Processing Module – Checks availability and manages logic

Storage Module – Stores data using ArrayList

Output Module – Displays results

Check-Out Module – Removes reservation

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Hotel Room Reservation and Guest Management System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis• System Design• Program Development• Testing and Validation• Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of restaurant operations.

Manual reservation systems are inefficient and error-prone. Need for automation.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

System designed with input, processing, and output modules using JavaFX.

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the **Java programming language and JavaFX**

The following steps were followed during development:

- UI creation
- Event handling
- Logic implementation

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Tested with:

- Valid inputs
- Duplicate room booking
- Check-out functionality

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented.

System works correctly and produces expected outputs.

The results are demonstrated through program execution and screenshots showing the input and output of the system.

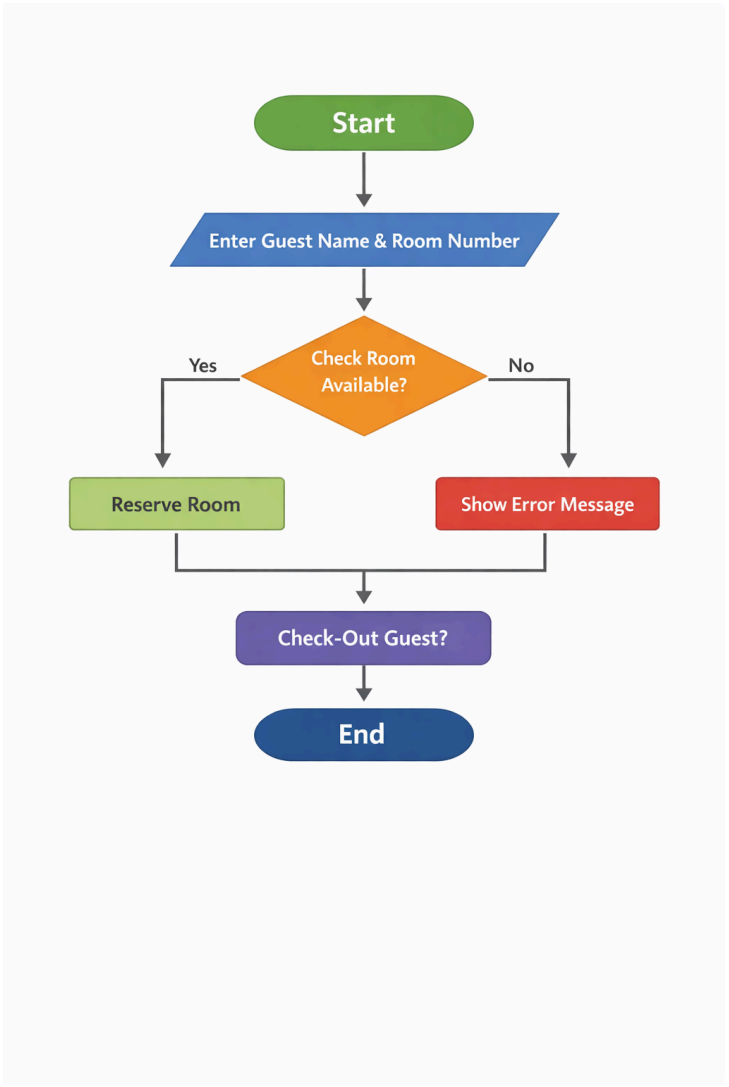
Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the **Hotel Room Reservation and Guest Management System**:

1. Start
2. Enter guest name and room number
3. Check room availability
4. If available → reserve room
5. Else → show error
6. Display reservations
7. Check-out guest if required
8. End

4.7. Flowchart


Start
Enter Guest Name & Room Number
Check Room Available?
Yes
No
Reserve Room
Show Error Message
Check-Out Guest?
End

5. SOURCE CODE

```
    // View Reservations
    viewBtn.setOnAction(e -> {
        displayArea.appendText("\n--- Current Reservations ---\n");
        if (reservations.isEmpty()) {
            displayArea.appendText("No active reservations\n");
        } else {
            for (String r : reservations) {
                displayArea.appendText(r + "\n");
            }
        }
    });

    // Check-Out Guest
    checkoutBtn.setOnAction(e -> {
        String name = guestName.getText();

        if (name.isEmpty()) {
            displayArea.appendText("Enter guest name to check out\n");
            return;
        }

        boolean found = false;

        for (int i = 0; i < reservations.size(); i++) {
            if (reservations.get(i).contains(name)) {
                reservations.remove(i);
                displayArea.appendText(name + " has checked out\n");
                found = true;
                break;
            }
        }

        if (!found) {
            displayArea.appendText("Guest not found\n");
        }

        guestName.clear();
    });
```

```
    // Reserve Room
    reserveRoomBtn.setOnAction(e -> {
String name = guestName.getText();
String room = roomNumber.getText();

if (name.isEmpty() || room.isEmpty()) {
    displayArea.appendText("Please enter both name and room number\n");
    return;
}

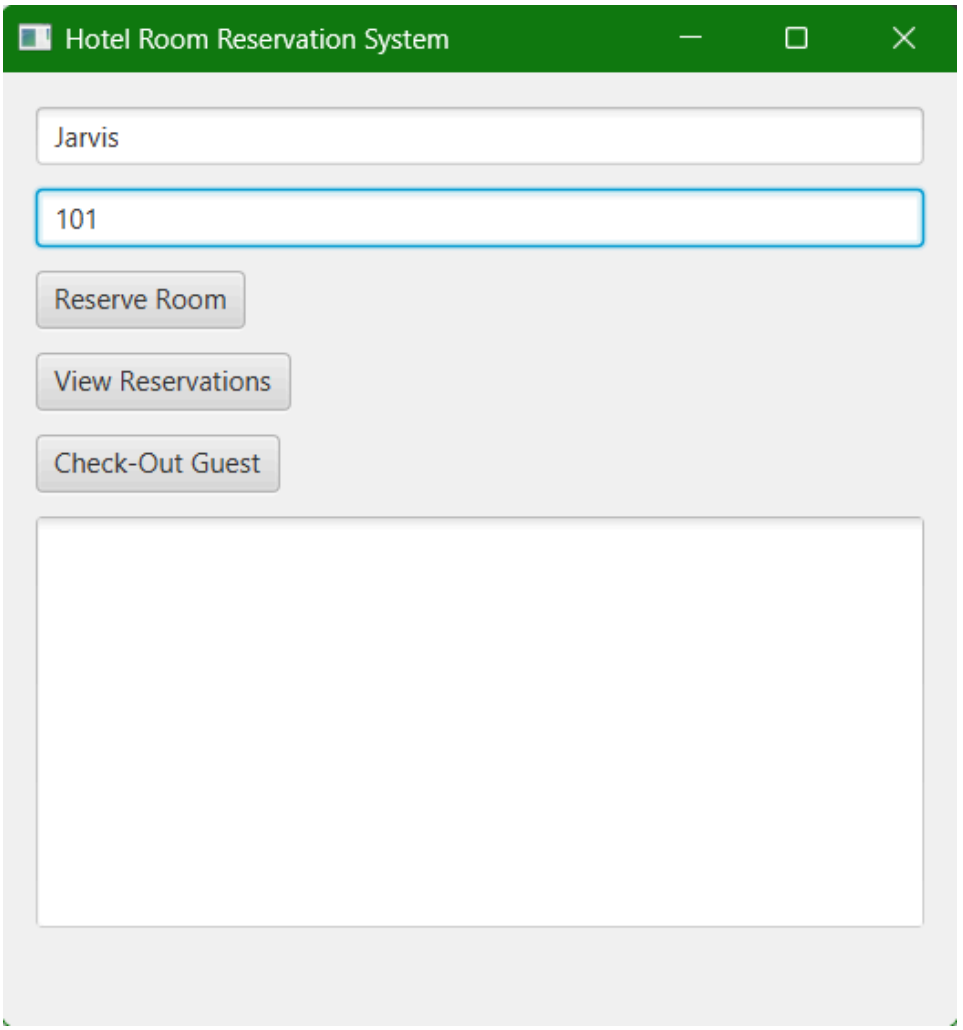
boolean roomBooked = false;

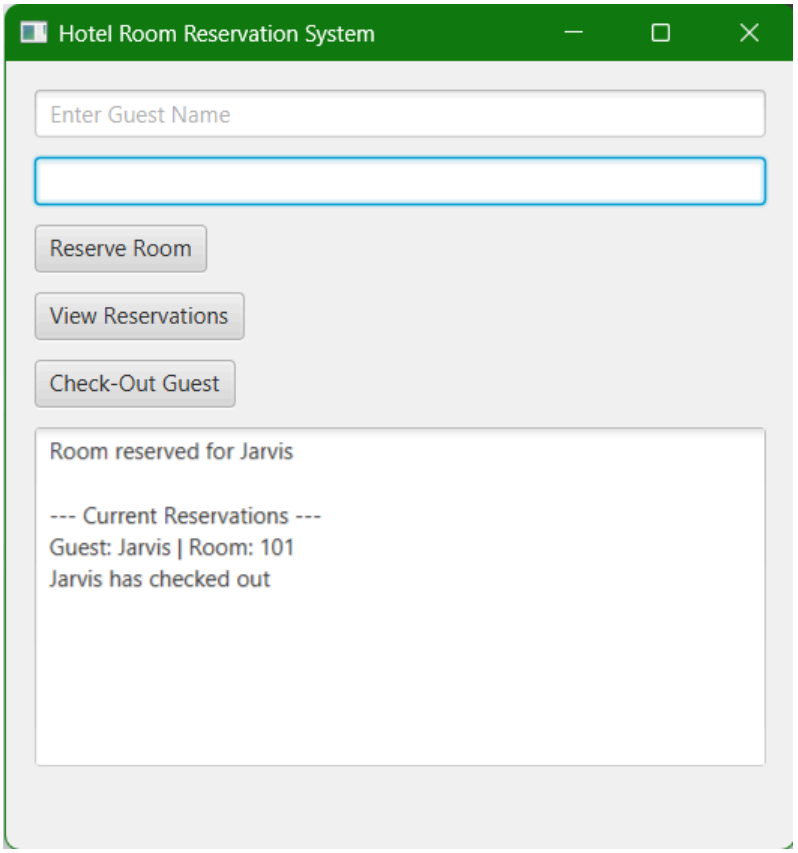
for (String r : reservations) {
    if (r.contains("Room: " + room)) {
        roomBooked = true;
        break;
    }
}

if (roomBooked) {
    displayArea.appendText("Room " + room + " is already booked\n");
} else {
    String record = "Guest: " + name + " | Room: " + room;
    reservations.add(record);
    displayArea.appendText("Room reserved for " + name + "\n");
}

guestName.clear();
roomNumber.clear();
});
```

6. INPUT AND OUTPUT SCREENS




Hotel Room Reservation System
Enter Guest Name
Reserve Room
View Reservations
Check-Out Guest
Room reserved for Jarvis
--- Current Reservations ---
Guest: Jarvis | Room: 101
Jarvis has checked out

7. DISCUSSION & CONCLUSION

7.1 Discussion

The system was developed using JavaFX to automate hotel reservations. It successfully manages bookings, prevents double booking, and handles guest check-out efficiently.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, and object-oriented principles were applied.

The system was tested with different inputs to ensure that it performs the required operations correctly. The results show that the system reduces manual effort and improves booking rooms efficiently.

7.2 Conclusion

The **Hotel Room Reservation and Guest Management System** successfully demonstrates how JavaFX can be used to develop a simple hotel management system. It improves efficiency and reduces errors in reservation handling.

The system automates the process of reserving rooms, reduces manual entry errors, and provides faster service to customers. It simplifies reserving a hotel room operations by managing the room and guest entries.

The project can be further enhanced by adding features such as database connectivity, graphical user interfaces, and online payment systems.

8. REFERENCES

The following resources were used during the development of this project:

- **Java: The Complete Reference – Herbert Schildt**
- **Oracle Java Documentation**
- **GeeksforGeeks**
- **W3Schools**